



(Credit: www.cptgroup.cn)

risk. Alternatively, if you are comfortable, it may be of low risk.

3. Has the solution considered all scenarios?

Following the premise of the solution that is designed right—one of the critical factors in evaluating the risks of the AI solution is the depth and breadth of design. You should understand and be confident that the solution at hand is fully equipped to handle the entire spectrum of scenarios that may occur in your business.

The AI systems do not perform their tasks consciously. These rely on the training data. It means that the system's reliability and reliability of its outcomes are at risk if the input data had problems. These data problems are not just bias; they are also related to the representation of various scenarios in the data.

Typical IT solutions, including AI and alike, are mostly point-based. These focus on one goal with a handful of other parameters to optimise when achieving that goal. Unfortunately, these solutions can never be all-inclusive and holistic; it is too much to expect from them.

Let us take the example of an autonomous car. If you ask the car to take you to your destination as fast as possible, it may get you there. But, while getting there, you may have broken several traffic rules and put fellow traveller's life at risk. None of this is acceptable. However, this may happen if your AI is too narrow and cannot see the forest of trees.

Many data with only a few types of scenarios cannot train the solution for real-life use cases, and this is a significant risk. **Unless your training dataset represents all the possible use cases and scenarios, it is useless.**

Since ML is an inductive process, the base model used by your solu-

tion can only cover what it has been trained on and seen in the data. If your training has missed low occurrence scenarios (also known as long-tail or edge cases), your solution will not be able to support them. Your solution will be too risky in those scenarios or may fail.

If the data is missing variety, it can lead to several problems in the future. The solution developed with a limited dataset and narrow training would be highly risky. Also, if the solution design has covered the breadth and depth of data and as many scenarios as reasonably possible, risks are lower.

4. Is there an option to stop gracefully?

A system that is designed right will always have an inbuilt redundancy. New-age AI solutions can be highly potent and may result in scalable damage when something goes wrong.

So, ask your development team, or your AI vendors, about the kill-switch. Check with them the possibility of stopping the system gracefully when needed.

A system without an option to stop gracefully is a highly risky system. You may not have an option to stop the system when needed. However, exercising this option is not as simple as pulling the plug. This means you may still face a significant risk of jeopardising business continuity in the process if you decide to stop it.

The ability to stop is not as bad as not being able to stop at all, though. In this case, you must evaluate your options and alternate plan. What are the financial and overall business repercussions of those alternative options when invoked? If stopping the AI system in an emergency would mean you are required to put up a hundred-member team in place to cope with the situation, it could be a

significant financial and HR burden for you.

When evaluating your AI solution on this parameter, check the availability of kill switch or graceful-stop and restart options. If there are no such options, it is a high-risk solution. However, if these options are available with some level of inconvenience and minor losses, it may be termed as a medium-risk solution. Although highly unlikely, there may be a solution that is easy to stop and restart. If it will not cost you too many resources, it may be a low-risk option and should be the most preferred one from a risk point of view.

If you cannot stop an AI solution easily and gracefully, it is a high-risk solution. On the contrary, if you are in full control and can exercise that option easily and without any significant loss, it is low risk, that is, a better solution.

5. Is the solution explainable and auditable?

A software solution, no matter how sophisticated, is always auditable, understandable, and explainable.

Specific to the AI solution, explainability has been one of the most talked-about topics. With so many different approaches to develop AI solutions, it is getting increasingly difficult to determine how systems are making decisions. With explainable and auditable AI solutions, we should be able to fix accountability, make corrections, and take corrective actions as well.

For many industries, AI explainability is a regulatory requirement. For example, with the General Data Protection Regulation (GDPR) laws in effect, companies will be required to provide consumers with an explanation for AI-based decisions.

There are growing risks of AI when implemented without adequate explainability. These risks affect ethics, business performance, regulation, and our ability to learn iteratively from AI implementations.

People do not want to accept arbitrary decisions made by some entity that they don't understand. To be acceptable and trustworthy, you would want to know why the system is making a specific decision.

In addition to explainability, being auditable is another key to the right AI solution. Whenever something goes